

List of Symbols

EART22001 Environmental Modelling

The symbols below summarise the notation used in the pressure/temperature/humidity/particulate and dynamical-models notes. Where a symbol is used in more than one context, the relevant meaning is stated.

Symbol	Meaning
A	Area (m^2), used in the pressure-force example.
$\mathbf{a}$	Acceleration vector (m s^{-2}).
β_{ext}	Light-extinction coefficient (m^{-1}).
c	Gravity-wave phase speed (m s^{-1}).
C_m	Particulate mass concentration (kg m^{-3} or $\mu\text{g m}^{-3}$).
D	Aerosol particle diameter (m).
e	Water-vapour partial pressure (Pa).
e_s	Saturation vapour pressure over liquid water (Pa).
ϵ	Ratio R_a/R_v , approximately 0.622.
f	Coriolis parameter (s^{-1}).
$\mathbf{F}$	Force vector (N).
$\mathbf{F}_A, \mathbf{F}_B$	Equal-and-opposite interaction forces in the Newton's-third-law example (N).
g	Acceleration due to gravity, approximately 9.81 m s^{-2} .
h	Mean fluid depth (m).
K	Koschmieder-type contrast constant; $K = 1.9$ is used in the visibility exercises.
L_v	Latent heat of vaporisation of water, approximately $2.5 \times 10^6 \text{ J kg}^{-1}$.
m	Mass (kg).
m_p	Mass of one spherical aerosol particle (kg).
M_a	Molar mass of dry air, approximately $28.97 \times 10^{-3} \text{ kg mol}^{-1}$.
M_s	Molar mass of dry aerosol solute (kg mol^{-1}).
M_w	Molar mass of water, $18 \times 10^{-3} \text{ kg mol}^{-1}$.
n	Amount of gas (mol).
n_s	Amount of dissolved solute in an aerosol particle (mol).
n_w	Amount of water in an aerosol particle (mol).
N_A	Avogadro constant, $6.022 \times 10^{23} \text{ mol}^{-1}$.
N_p	Aerosol particle number concentration (m^{-3}).
ν	Effective dissociation (van't Hoff) factor.
Ω	Earth's angular rotation rate, approximately $7.292 \times 10^{-5} \text{ s}^{-1}$.
P	Pressure (Pa).
ΔP	Pressure difference (Pa).
PGF_{vol}	Pressure-gradient force per unit volume (N m^{-3}).
ϕ	Latitude.
Q_{ext}	Dimensionless light-extinction efficiency.
R	Radius or radius of curvature (m).
R_a	Specific gas constant for dry air, approximately $287 \text{ J kg}^{-1} \text{ K}^{-1}$.
R_{gas}	Universal gas constant, $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.
R_v	Specific gas constant for water vapour, approximately $461 \text{ J kg}^{-1} \text{ K}^{-1}$.
RH	Relative humidity, expressed as a fraction unless stated otherwise.
r_v	Water-vapour mass mixing ratio (kg kg^{-1} of dry air).
ρ	Fluid/air density (kg m^{-3}) in the dynamical-model equations.
ρ_a	Density of air (kg m^{-3}).
ρ_d	Density of dry air (kg m^{-3}).
ρ_p	Particle density (kg m^{-3}).
ρ_s	Density of dry aerosol solute (kg m^{-3}).
ρ_{sol}	Density of the water-solute solution (kg m^{-3}).
ρ_v	Water-vapour density (kg m^{-3}).
ρ_w	Density of liquid water, approximately 1000 kg m^{-3} .
σ	Surface tension of water; approximately 0.075 N m^{-1} in the exercises.
t	Time (s).
T	Absolute temperature (K); in the circular-motion discussion, T is also used locally for the period of one revolution.
u, v	Eastward and northward velocity components (m s^{-1}).
u_g, v_g	Components of the geostrophic wind (m s^{-1}).

Symbol	Meaning
v_{gr}	Gradient-wind speed (m s^{-1}).
v_t	Tangential speed in circular motion (m s^{-1}).
V	Gas volume (m^3).
V_{vis}	Visual range (m).
$\mathbf{v}$	Velocity vector (m s^{-1}).
W	Weight (N).
x, y	Eastward and northward position coordinates (m).
Δx	Horizontal distance interval (m).
$\ddot{x}, \ddot{y}$	Acceleration components obtained as second time derivatives of position.
ω	Angular frequency (rad s^{-1}).
ζ	Vertical component of relative vorticity (s^{-1}).